Note 3.2 The Growth of Functions

Introduction

In this section we will estimate the time of an algorithm, the growth of a function to solve the problem and show the methods.

Big-O Notation

Definition of Big-O Notation

Let f and g be functions from the set of integers or real numbers to the set of real numbers. Then we say $f(x)$ is $O(g(x))$ if there are constants C and k such that

$$|f(x)| \leq C\,|g(x)|$$

where $x > k$. The constants C and k are called witnesses to the relationship. (Obviously C is positive). So when proving, we just need to find the pair of witnesses.

Intuitively, the definition says that $f(x)$ grows slower or equivalent than some fixed multiple of $g(x)$ as x grows without bound. That is, the value of $f(x)$ is always be limited by the fixed multiple of $g(x)$ when $x \to \infty$.

Sometimes we express it as

$$f(x) = O(g(x))$$

but the equals sign is not genuine equality. Actually $O(g(x))$ is a set and the sign represents in the set.

Working with the Definition of Big-O Notation

A useful approach for finding a pair of witnesses is to first select a value of k for which the size of $|f(x)|$ can be readily estimated when $x > k$ and to see whether to find a constant C satisfy the definition.

To prove one function is not the big-O of another, we can prove by the contradiction that suppose that there exist k, C.

And When using big-O notation, the function g in the big-O is chosen to be as small as possible.

The Same Order

If the two function f and g can be expressed as mutual big-O of each other, that is

$$f(x)=O(g(x)), g(x)=O(f(x))$$

Then we can say they are of the same order.

Big-O Estimates for Some Important Functions

Theorem 1

Let $f(x)=a_nx^n+a_{n-1}x^{n-1}+\cdots+a_1x+a_0$, where $a_0, a_1, \cdots, a_n$ are real numbers. Then $f(x)$ is $O(x^n)$.

Important Functions

The next functions are sequenced in ascending:

$$1, \log n, n, n\log n, n^2, 2^n, n!$$

The Growth of Combinations of Functions

Theorem 2

Suppose that $f_1(x)$ is $O(g_1(x))$ and that $f_2(x)$ is $O(g_2(x))$. Then $(f_1+f_2)(x)$ is

$$O(\max(|g_1(x)|, |g_2(x)|))$$

Corollary 1

Suppose that $f_1(x), f_2(x)$ are both $O(g(x))$, then $(f_1+f_2)(x)$ is $O(g(x))$.

Theorem 3

Suppose that $f_1(x)$ is $O(g_1(x))$ and $f_2(x)$ is $O(g_2(x))$, then $(f_1f_2)(x)$ is $O(g_1(x)g_2(x))$.

Big-Omega and Big-Theta Notation

Definition of Big-Omega(Ω) Notation

Let f and g be functions from the set of integers or the set of real numbers to the set of real numbers. We say $f(x)$ is $\Omega(g(x))$ if there are positive constants C and k such that

$$|f(x)| \geq C\,|g(x)|$$

whenever $x>k$.

We can know that

$$f(x) \text{ is } \Omega(g(x)) \Leftrightarrow g(x) \text{ is } O(f(x))$$

Definition of Big-Theta(Θ) Notation

Let f and g be functions from the set of integers or the set of real numbers to the set of real numbers. We say that $f(x)$ is $\Theta(g(x))$ if $f(x)$ is $O(g(x))$ and $f(x)$ is $\Omega(g(x))$. That is $f(x)$ and $g(x)$ are of the same order. (or $f(x)$ is of order $g(x)$).

Theorem 4

Let $f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$, where the coefficients are real numbers with $a_n \neq 0$. Then $f(x)$ is of order x^n.